



UCCD2203 Database Systems

Lecture 4

By

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Lecture 4 Topic: Database Design Methodology - Part 2

- Physical Database Design
- Advance Normalization
- De-Normalization

PHYSICAL DATABASE DESIGN

Comparison of Logical and Physical Database Design

- Sources of information for physical design process includes logical data model and documentation that describes model.
- **Logical database design** is concerned with the *what*,
- **Physical database design** is concerned with the *how*.

Design Representation of Derived Data

Objective: To decide how to represent any derived data present in the global logical data model in the target DBMS.

- Examine logical data model and data dictionary, and produce list of all derived attributes.
- Derived attribute can be stored in database or calculated every time it is needed.

PropertyForRent relation and Staff relation with derived attribute noOfProperties

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

Staff

staffNo	fName	lName	branchNo	noOfProperties
SL21	John	White	B005	0
SG37	Ann	Beech	B003	2
SG14	David	Ford	B003	1
SA9	Mary	Howe	B007	1
SG5	Susan	Brand	B003	0
SL41	Julie	Lee	B005	1

Design general Constraints

Objective: To design the general constraints for the target DBMS.

- Some DBMS provide more facilities than others for defining general constraints. Example:

```
CONSTRAINT StaffNotHandlingTooMuch  
CHECK (NOT EXISTS (SELECT staffNo  
                    FROM PropertyForRent  
                    GROUP BY staffNo  
                    HAVING COUNT(*) > 100))
```

Design File Organization and Indexes

- **Objective:** To determine optimal file organizations to store the base relations and the indexes that are required to achieve acceptable performance; that is, the way in which relations and tuples will be held on secondary storage.
- The activities in Step 4:
 - Step 4.1 Analyze transactions
 - Step 4.2 Choose file organizations
 - Step 4.3 Choose indexes
 - Step 4.4 Estimate disk space requirements

Analyze Transactions

- **Objective:** To understand the **functionality** of the transactions that will run on the database and to analyze the important transactions.
- Attempt to identify performance criteria, such as:
 - ❑ transactions that run **frequently** and will have a significant impact on performance;
 - ❑ transactions that are **critical** to the business;
 - ❑ times during the **day/week** when there will be a high demand made on the database (called the *peak load*).

Cross-referencing transactions and relations

Table 16.1 Cross-referencing transactions and relations.

Transaction/ Relation	(A)				(B)				(C)				(D)				(E)				(F)			
	I	R	U	D	I	R	U	D	I	R	U	D	I	R	U	D	I	R	U	D	I	R	U	D
Branch									X				X								X			
Telephone																								
Staff		X				X			X								X				X			
Manager																								
PrivateOwner	X																							
BusinessOwner	X																							
PropertyForRent	X					X	X	X					X				X				X			
Viewing																								
Client																								
Registration																								
Lease																								
Newspaper																								
Advert																								

I = Insert; R = Read; U = Update; D = Delete

Choose File Organizations

- **Objective:** To determine an efficient file organization for each base relation.
- File organizations include
 - Heap,
 - Hash,
 - Indexed Sequential Access Method (ISAM),
 - B+-Tree, and
 - Clusters.

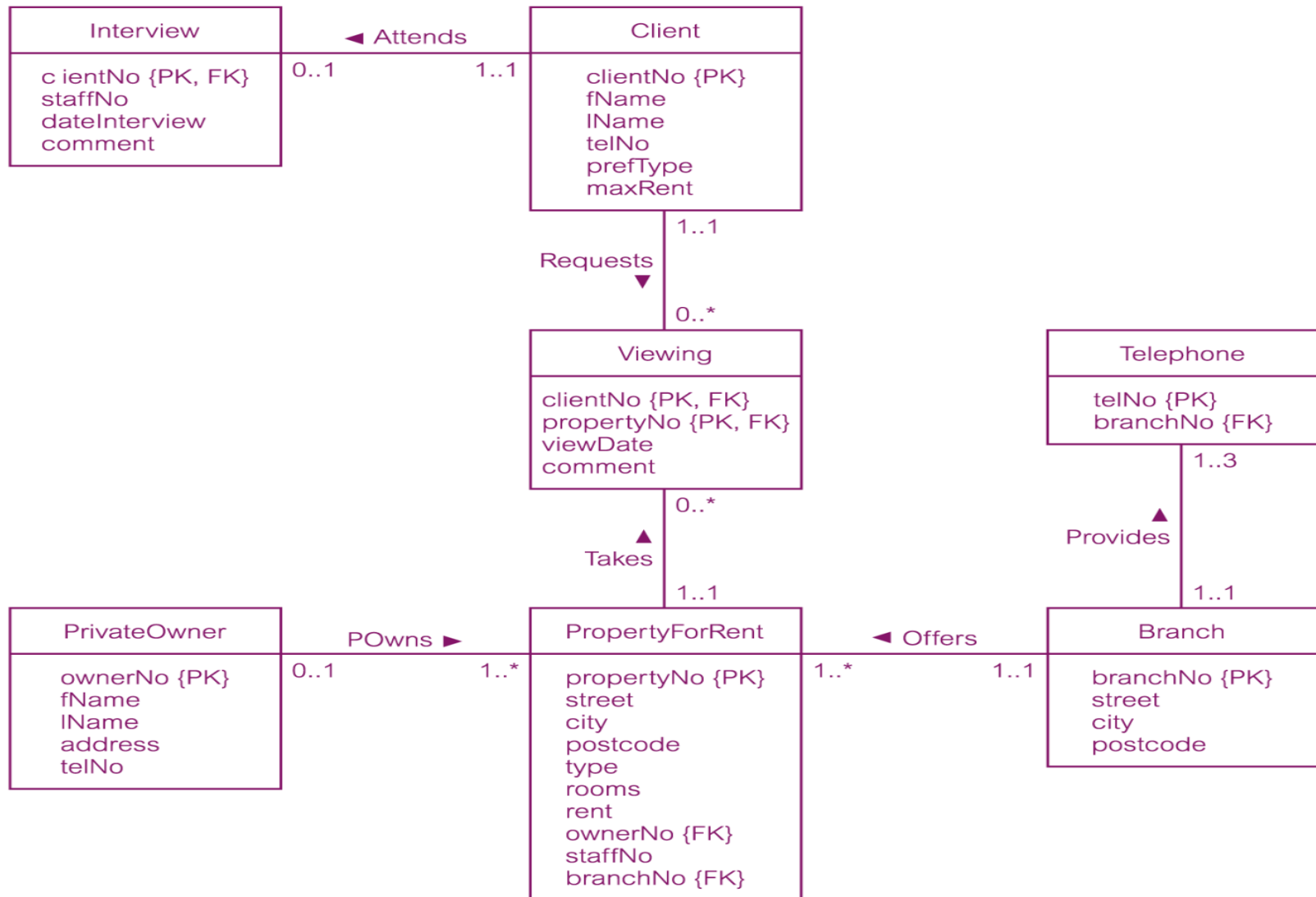
Choose Indexes

- **Objective:** To determine whether adding indexes will improve the **performance** of the system.
- One approach is to **keep tuples unordered** and create as many secondary indexes as necessary.

Note: Indexes are special **lookup tables** that the database search engine can use to **speed up data retrieval**.

Simply put, an index is a pointer to data in a table. An index in a database is very similar to an index in the back of a **book**.

Sample Relation Diagram



Sample Relations

Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Telephone

telNo	branchNo
0207-886-1212	B005
0207-886-1300	B005
0207-886-4100	B005
01224-67125	B007
0141-339-2178	B003
0141-339-4439	B003
0117-916-1170	B004
0208-963-1030	B002

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

Client

clientNo	fName	lName	telNo	prefType	maxRent
CR76	John	Kay	0207-774-5632	Flat	425
CR56	Aline	Stewart	0141-848-1825	Flat	350
CR74	Mike	Ritchie	01475-392178	House	750
CR62	Mary	Tregear	01224-196720	Flat	600

Interview

clientNo	staffNo	dateInterview	comment
CR56	SG37	11-Apr-00	current lease ends in June
CR62	SA9	7-Mar-00	needs property urgently

PrivateOwner

ownerNo	fName	lName	address	telNo
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025

Viewing

clientNo	propertyNo	viewDate	comment
CR56	PA14	24-May-01	too small
CR76	PG4	20-Apr-01	too remote
CR56	PG4	26-May-01	
CR62	PA14	14-May-01	
CR56	PG36	28-Apr-01	no dining room

(b)

Combining 1:1 relationships



(a)

ClientInterview

clientNo	fName	lName	telNo	prefType	maxRent	staffNo	dateInterview	comment
CR76	John	Kay	0207-774-5632	Flat	425			
CR56	Aline	Stewart	0141-848-1825	Flat	350	SG37	11-Apr-03	current lease ends in June
CR74	Mike	Ritchie	01475-392178	House	750			
CR62	Mary	Tregear	01224-196720	Flat	600	SA9	7-Mar-03	needs property urgently

(b)

Figure 18.2 Combined Client and Interview: (a) revised extract from the relation diagram; (b) combined relation.

Duplicating non-key attributes in 1:* relationships to reduce joins

PropertyForRent

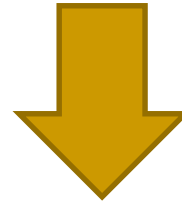


propertyNo	street	city	postcode	type	rooms	rent	ownerNo	IName	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	Keogh	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	Farrel	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40	Murphy		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	Shaw	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	Farrel	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	Shaw	SG14	B003

Duplicating non-key attributes in 1:* relationships: Lookup Table

PropertyType

type	description
H	House
F	Flat



PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	H	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	F	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	F	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	F	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	H	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	F	4	450	CO93	SG14	B003

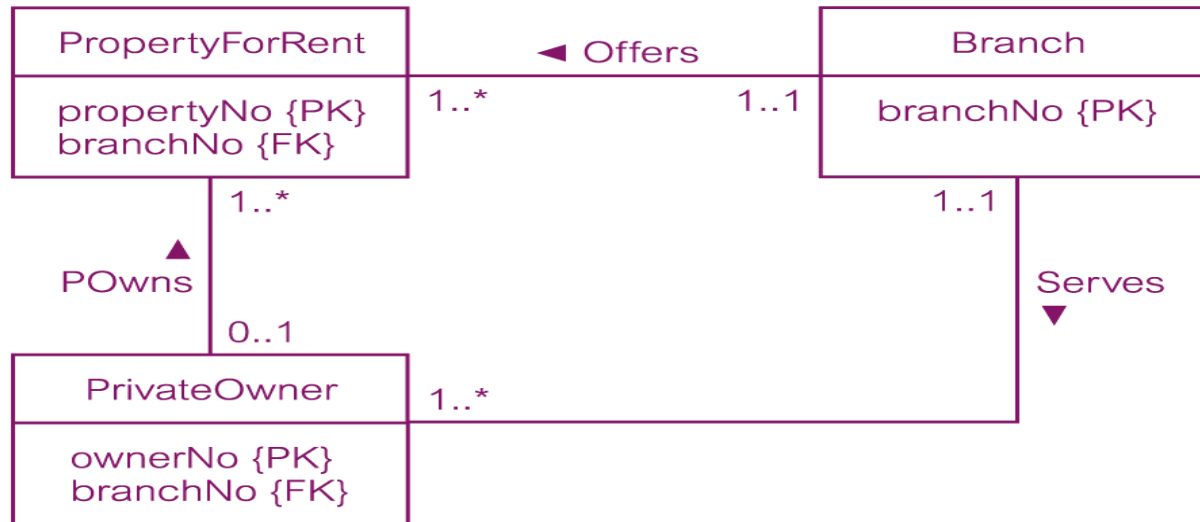
Duplicating non-key attributes in 1:* relationships: Lookup Table

PropertyForRent

propertyNo	street	city	postcode	type	description	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	H	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	F	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	F	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	F	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	H	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	F	Flat	4	450	CO93	SG14	B003

Figure 18.5 Modified PropertyForRent relation with duplicated description attribute.

Duplicating FK attributes in 1:* relationship to reduce joins



(a)

PrivateOwner

ownerNo	fName	lName	address	telNo	branchNo
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212	B007
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419	B003
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728	B003
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025	B003

(b)

Duplicating attributes in *:~ relationships to reduce joins

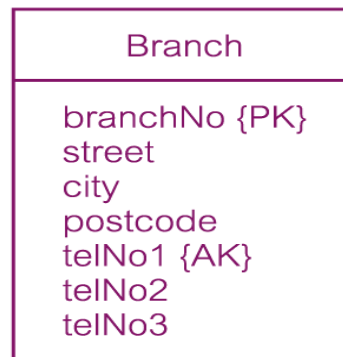
Duplicating the street attribute from the **PropertyForRent** relation in the Viewing relation

Viewing

clientNo	propertyNo	street	viewDate	comment
CR56	PA14	16 Holhead	24-May-04	too small
CR76	PG4	6 Lawrence St	20-Apr-04	too remote
CR56	PG4	6 Lawrence St	26-May-04	
CR62	PA14	16 Holhead	14-May-04	no dining room
CR56	PG36	2 Manor Rd	28-Apr-04	

Introducing repeating groups

Branch incorporating repeating group: (a) revised relation diagram, (b) revised relation



(a)

Branch

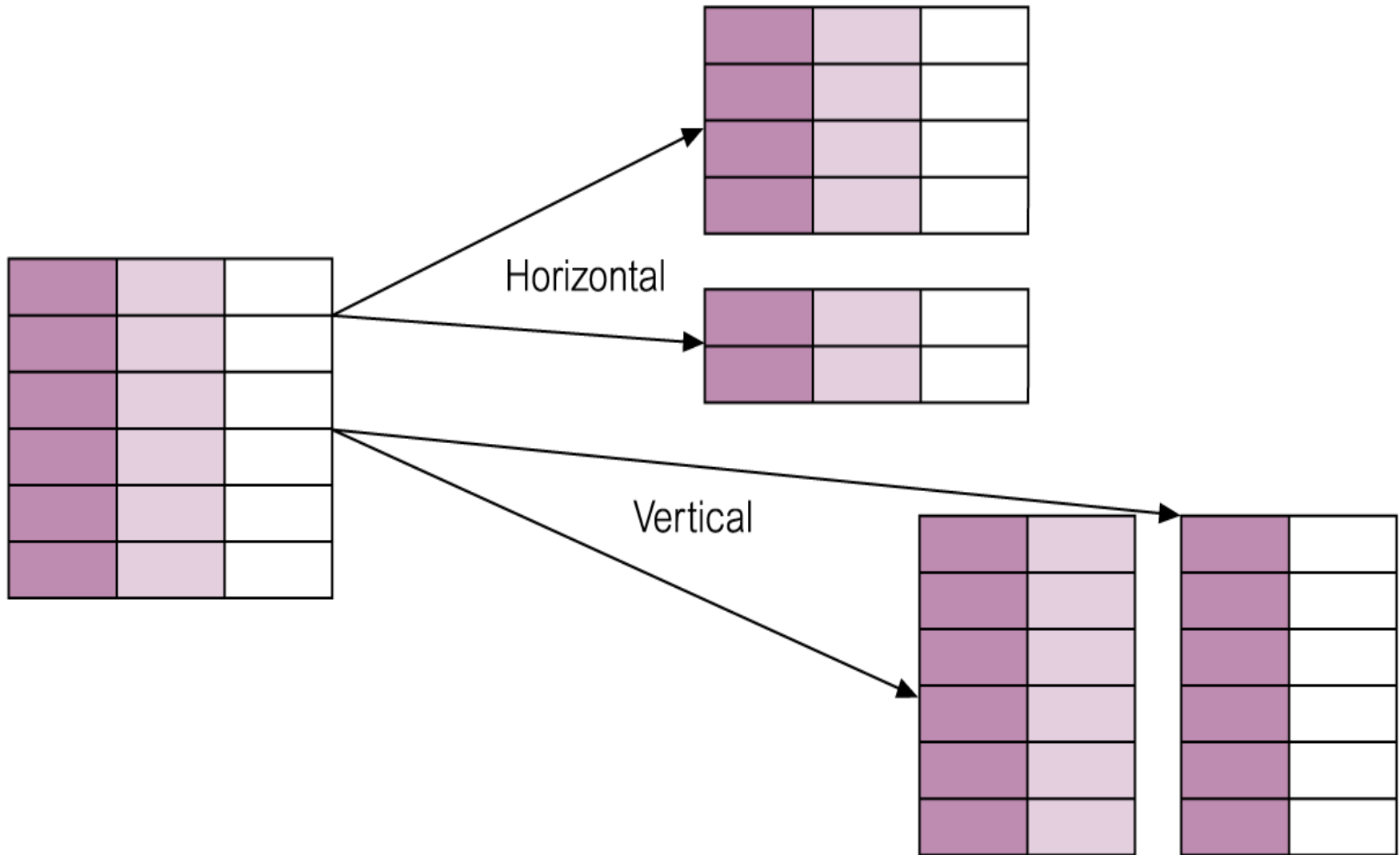
branchNo	street	city	postcode	telNo1	telNo2	telNo3
B005	22 Deer Rd	London	SW1 4EH	0207-886-1212	0141-339-4439	0207-886-4100
B007	16 Argyll St	Aberdeen	AB2 3SU	01224-67125		
B003	163 Main St	Glasgow	G11 9QX	0141-339-2178		
B004	32 Manse Rd	Bristol	BS99 1NZ	0117-916-1170		
B002	56 Clover Dr	London	NW10 6EU	0208-963-1030		

(b)

Partitioning relations

- Rather than combining relations together, alternative approach is to decompose them into a number of smaller and more manageable partitions.
- Two main types of partitioning: horizontal and vertical.

Partitioning relations



Monitor and tune the operational system

- **Objective:** To monitor the operational system and improve the performance of the system to correct inappropriate design decision or reflect changing requirements.
- Number of factors that used to measure efficiency:
 - Transaction throughput,
 - Response time,
 - Disk storage.

Monitor & Tune Operational System

- Number of factors may be used to measure efficiency:
 - *Transaction throughput*: number of transactions processed in given time interval.
 - *Response time*: elapsed time for completion of a single transaction.
 - *Disk storage*: amount of disk space required to store database files.

ADVANCE NORMALIZATION AND DE-NORMALIZATION

Data redundancy and Update Anomalies

- A relation should contains minimal data redundancy and allows users to insert, delete, and update rows without causing data inconsistencies
- The following anomalies to be avoided:
 - **Insertion Anomaly** – adding new rows forces user to create duplicate data
 - **Update or Modification Anomaly** – changing data in a row forces changes to other rows because of duplication
 - **Deletion Anomaly** – deleting rows may cause a loss of data that would be needed for other future rows

Insertion Anomaly

Insertion Anomaly – adding new rows forces user to create duplicate data

Faculty and Their Courses

Faculty ID	Faculty Name	Faculty Hire Date	Course Code
389	Dr. Giddens	10-Feb-1985	ENG-206
407	Dr. Saperstein	19-Apr-1999	CMP-101
407	Dr. Saperstein	19-Apr-1999	CMP-201

424

Dr. Newsome

29-Mar-2007

?

INSERT the same record again in another relation

424

Dr. Newsome

29-Mar-2007

Update or Modification Anomaly

Update or Modification Anomaly – changing data in a row forces changes to other rows because of duplication

Employees' Skills

Employee ID	Employee Address	Skill
426	87 Sycamore Grove	Typing
426	87 Sycamore Grove	Shorthand
519	94 Chestnut Street	Public Speaking
519	96 Walnut Avenue	Carpentry

Deletion Anomaly

Deletion Anomaly – deleting rows may cause a loss of data that would be needed for other future rows

Faculty and Their Courses

Faculty ID	Faculty Name	Faculty Hire Date	Course Code
389	Dr. Giddens	10-Feb-1985	ENG-206
407	Dr. Saperstein	19-Apr-1999	CMP-101
407	Dr. Saperstein	19-Apr-1999	CMP-201

DELETE

INVOICE Data (Pine Valley Furniture Company)

<u>Order_ID</u>	<u>Order_</u> Date	<u>Customer_</u> ID	<u>Customer_</u> Name	<u>Customer_</u> Address	<u>Product_ID</u>	<u>Product_</u> Description	<u>Product_</u> Finish	<u>Unit_</u> Price	<u>Ordered_</u> Quantity
1006	10/24/2006	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
1006	10/24/2006	2	Value Furniture	Plano, TX	5	Writer's Desk	Cherry	325.00	2
1006	10/24/2006	2	Value Furniture	Plano, TX	4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2006	6	Furniture Gallery	Boulder, CO	11	4-Dr Dresser	Oak	500.00	4
1007	10/25/2006	6	Furniture Gallery	Boulder, CO	4	Entertainment Center	Natural Maple	650.00	3

Anomalies in this Table

Refer to previous figure (*INVOICE Data (Pine Valley Furniture Company)*):

- **Insertion** – if new product is ordered for **Order 1007** of existing customer, customer data must be re-entered, causing duplication.
- **Deletion** – if we delete the **Dining Table** from **Order 1006**, we lose information concerning this item's finish and price.
- **Modification** – changing the price of **product ID 4** requires update in several records.

Why do these anomalies exist?

- Because there are multiple themes (**entity types**) into one relation. This results in duplication, and an unnecessary dependency between the entities.

Data Normalization

- **Normalization**: A technique for producing a set of relations with desirable properties.
- Primarily a process to validate and improve a logical design so that it satisfies certain constraints that **avoid unnecessary duplication of data**.
- The process of decomposing relations with anomalies to produce **smaller, *well-structured relations***.
- **Anomalies can be removed** by splitting the relation into two or more relations; each with a different, single theme.
- However, breaking up a relation may create referential integrity constraints.
- Normalization works through classes of relations called normal forms.

Functional Dependencies and Keys

Functional Dependency: The value of one attribute (**the determinant**) determines the value of another attribute.

- If A and B are attributes of relation R, B is functionally dependent on A (**denoted $A \rightarrow B$**), if each value of A in R is associated with exactly one value of B in R.

Determinant: Refers to the attribute, or group of attributes, on the left-hand side of the arrow of a functional dependency.

Candidate Key: A unique identifier. One of the candidate keys will become the primary key.

- E.g. perhaps there is both credit card number and SSN (**social security number**) in a table, in this case both are candidate keys.
- Each non-key field is functionally dependent on every candidate key.

Functional Dependencies and Keys (Cont.)

Full functional dependency:

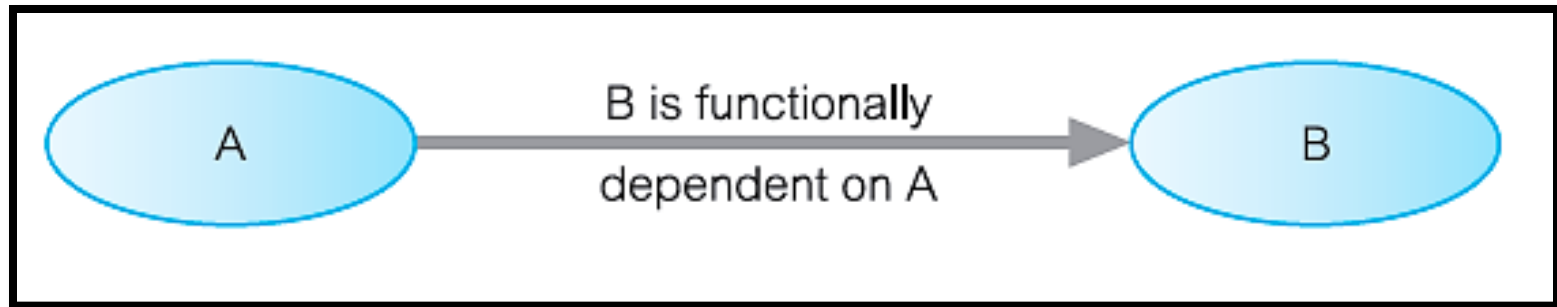
If attribute B is functionally dependent on a composite key A but not on any subset of that composite key, the attribute B is fully functionally dependent on A.

E.g. A functional dependency $A \rightarrow B$ is a full functional dependency if removal of any attribute from A results in the dependency no longer exists.

Partial Dependency: A functional dependency $A \rightarrow B$ is a partially dependency if there is some attribute that can be removed from A and yet the dependency still holds.

Transitive dependency: A condition where A, B, and C are attributes of a relation such that if $A \rightarrow B$ and $B \rightarrow C$, then C is transitively dependent on A via B (provided that A is not functionally dependent on B or C).

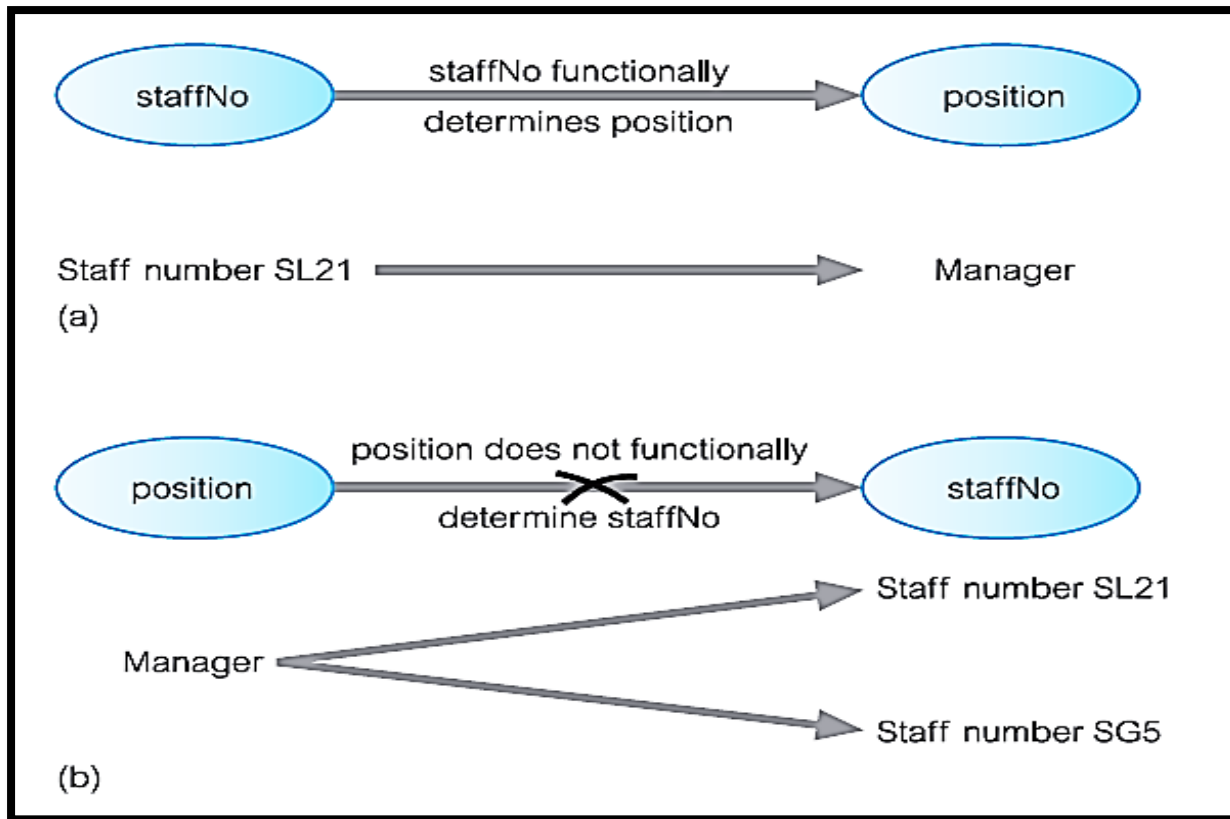
A Functional Dependency Diagram



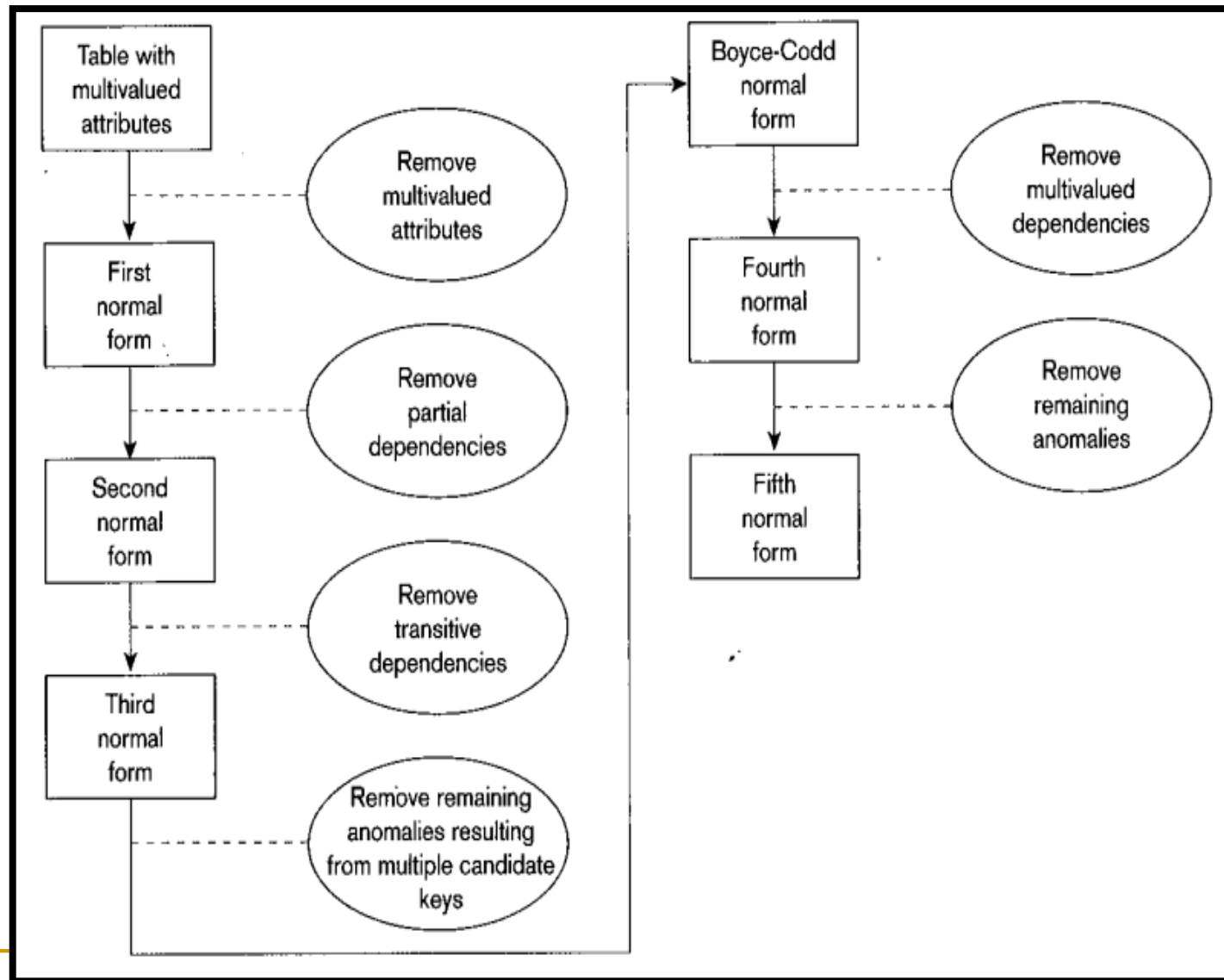
$A \twoheadrightarrow B$

A Functional Dependency Diagram

- (a) staffNo functionally determines position ($\text{staffNo} \rightarrow \text{position}$);
- (b) position does not functionally determine staffNo ($\text{position} \nrightarrow \text{staffNo}$).



Process of Normalization



Unnormalized Form (UNF)

Unnormalized Form (UNF): A table that contains one or more repeating groups.

We begin the process of normalization by first transferring the data from the source (for example, a standard data entry form) into table format with rows and columns.

In this format, the table is in Unnormalized Form and is referred to as an unnormalized table.

To transform the unnormalized table to First Normal Form we identify and remove repeating groups within the table.

There are two common approaches to removing repeating groups from unnormalized tables:

- By entering appropriate data in the empty columns of rows containing the repeating data.
- By placing the repeating data, along with a copy of the original key attribute(s), in a separate relation.

Branch table is not in 1NF

branchNo	branchAddress	telNos
B001	8 Jefferson Way, Portland, OR 97201	503-555-3618, 503-555-2727, 503-555-6534
B002	City Center Plaza, Seattle, WA 98122	206-555-6756, 206-555-8836
B003	14 – 8th Avenue, New York, NY 10012	212-371-3000
B004	16 – 14th Avenue, Seattle, WA 98128	206-555-3131, 206-555-4112

Primary key

More than
one value, so
not in 1NF

First Normal Form (1NF)

First Normal Form (1NF): A relation in which the intersection of each row and column contains one and only one value.

- No multivalued attributes, no repeating group, every attribute value is atomic.
- All *relations* are in 1st Normal Form.

Fig. A is not in 1st Normal Form (**Repeating group**) → it is not a relation.

Fig. B is in 1st Normal form.

Fig. A: Table with multivalued attributes, not in 1NF

INVOICE Data (Pine Valley Furniture Company)

<u>Order_ID</u>	Order_ Date	Customer_ ID	Customer_ Name	Customer_ Address	<u>Product_ID</u>	Product_ Description	Product_ Finish	Unit_ Price	Ordered Quantity
1006	10/24/2006	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
					5	Writer's Desk	Cherry	325.00	2
					4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2006	6	Furniture Gallery	Boulder, CO	11	4-Dr Dresser	Oak	500.00	4
					4	Entertainment Center	Natural Maple	650.00	3

Note: this is NOT a relation.

Fig. B: Table with no multivalued attributes and unique rows, is in 1NF

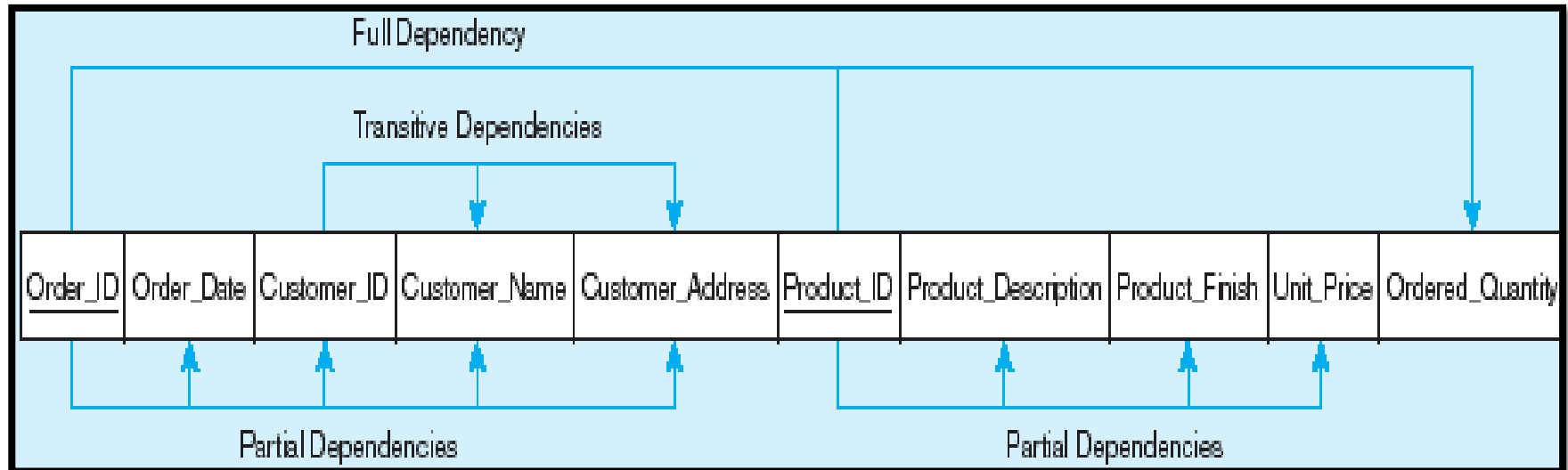
INVOICE Data (Pine Valley Furniture Company)

<u>Order_ID</u>	Order_ Date	Customer_ ID	Customer_ Name	Customer_ Address	<u>Product_ID</u>	Product_ Description	Product_ Finish	Unit_ Price	Ordered_ Quantity
1006	10/24/2006	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
1006	10/24/2006	2	Value Furniture	Plano, TX	5	Writer's Desk	Cherry	325.00	2
1006	10/24/2006	2	Value Furniture	Plano, TX	4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2006	6	Furniture Gallery	Boulder, CO	11	4-Dr Dresser	Oak	500.00	4
1007	10/25/2006	6	Furniture Gallery	Boulder, CO	4	Entertainment Center	Natural Maple	650.00	3

Note: this is relation in 1NF, but not a well-structured one

Functional Dependencies, in 1NF but Not in 2NF

INVOICE Data (Pine Valley Furniture Company)



Order ID, Product ID → Order_Date, Customer_ID, Customer_Name, Customer_Address, Product_Description, Product_Finish, Unit_Price, Order_Quantity

Therefore, NOT in 2NF

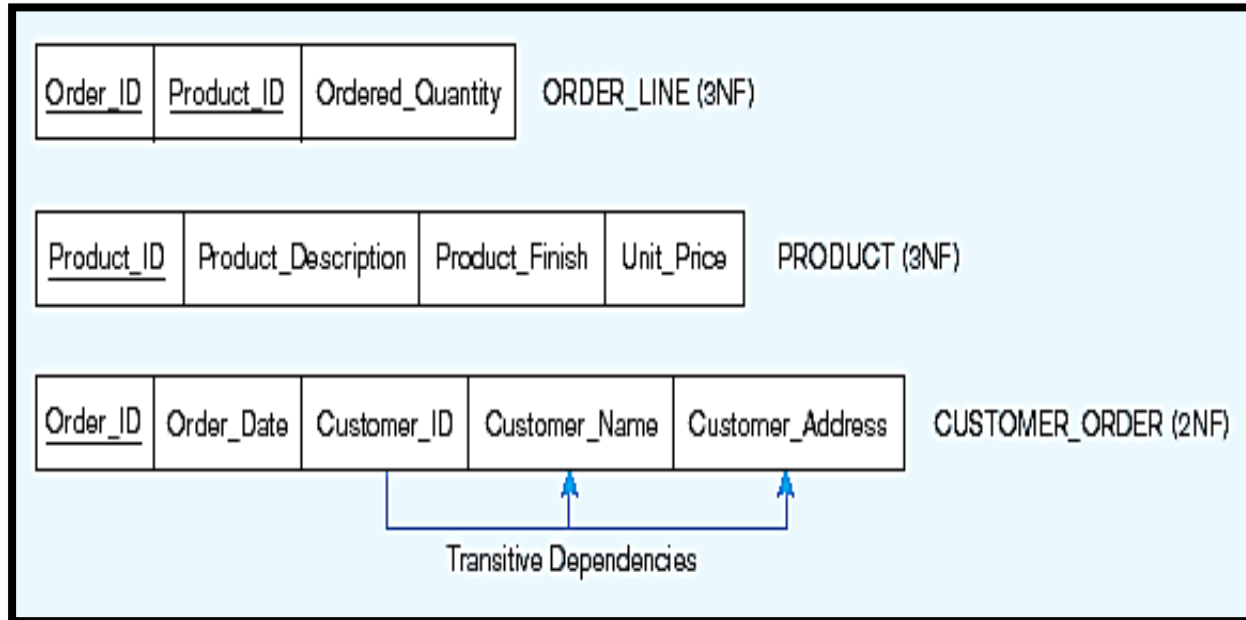
Second Normal Form (2NF)

Second Normal Form (2NF): A relation that is in First Normal Form and every non-primary-key attribute is fully functionally dependent on the primary key.

- Every non-key attribute must be defined by the entire key, not by only part of the key.
- No partial functional dependencies.

Functional Dependencies, 2NF but not in 3NF

INVOICE Data (Pine Valley Furniture Company)



Order_ID → Order_Date, Customer_ID, Customer_Name, Customer_Address
(Partial Dependencies)

Product_ID → Product_Description, Product_Finish, Unit_Price
(Partial Dependencies)

Order_ID, Product_ID → Order_Quantity
(Full Dependencies)

Transitive Dependencies still exist

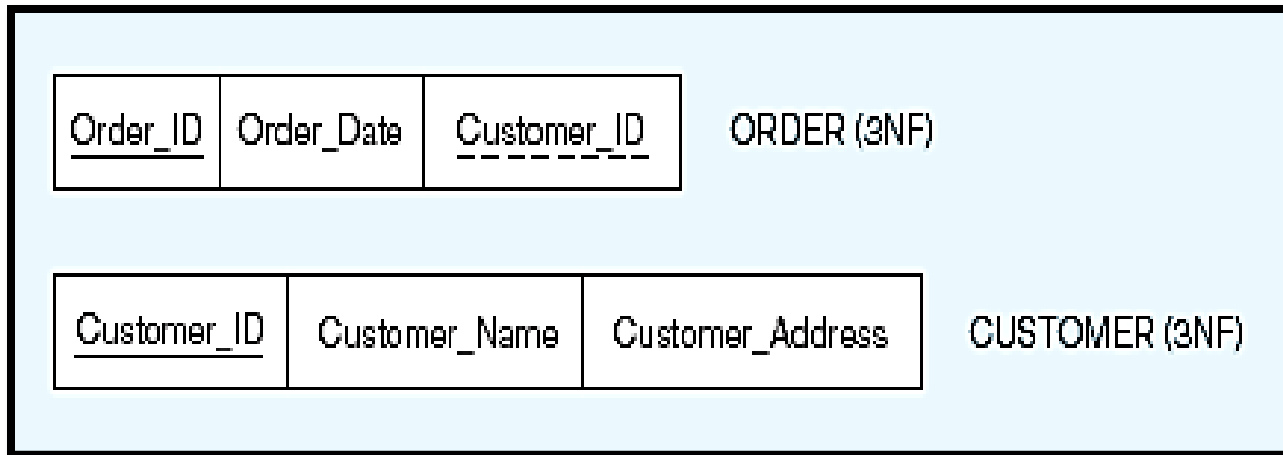
Third Normal Form (3NF)

Third Normal Form (3NF): A relation that is in First and Second Normal Form and in which no non-primary-key attribute is transitively dependent on the primary key.

- 2NF and no transitive dependencies (**functional dependencies on non-primary-key attributes**)
 - **Note**: this is called transitive, because the primary key is a determinant for another attribute, which in turn is a determinant for a third.
 - **Solution**: non-key determinant with transitive dependencies go into a new table; non-key determinant becomes primary key in the new table and stays as foreign key in the old table.

Functional Dependencies in 3NF

INVOICE Data (Pine Valley Furniture Company)



Order_ID → Order_Date, Customer_ID *
(Partial Dependencies)

Product_ID → Product_Description, Product_Finish, Unit_Price
(Partial Dependencies)

Order_ID, Product_ID → Order_Quantity
(Full Dependencies)

Customer_ID → Customer_Name, Customer_Address
(Transitive Dependencies)

Transitive dependencies are removed

Boyce-Codd Normal Form (BCNF)

- A relation is in **BCNF** if every determinant is a candidate key.
- For a relation with only one candidate key, **3NF** and **BCNF** are equivalent.
- Violation of **BCNF** is rare, may occur in a relation that contains two (or more) composite candidate keys and which overlap, that is share at least one attribute in common.
- Non primary key attribute determines part of the primary key attribute.

Converting to BCNF:

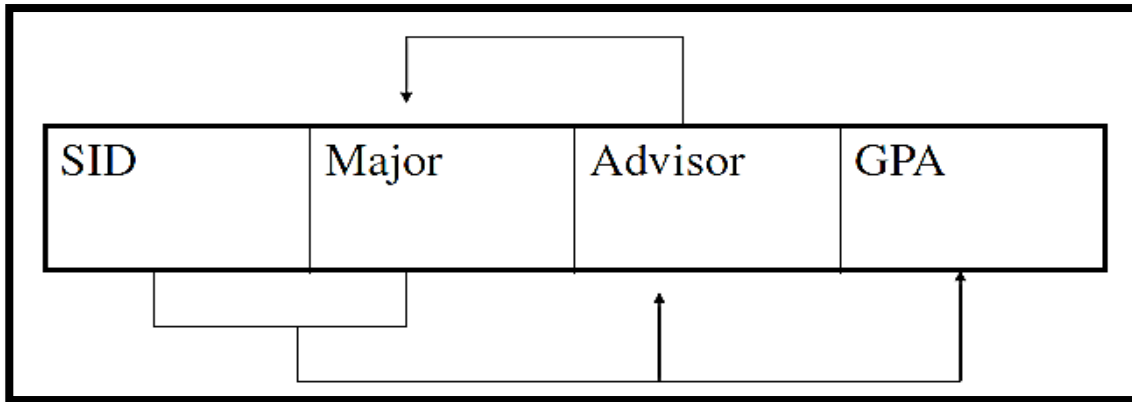
- **Step 1:** Make the determinant that is not a candidate key to become a component of the Primary Key of the revised table.
- **Step 2:** Decomposed the relation to eliminate the partial dependency.

Student_Advisor relation in 3NF

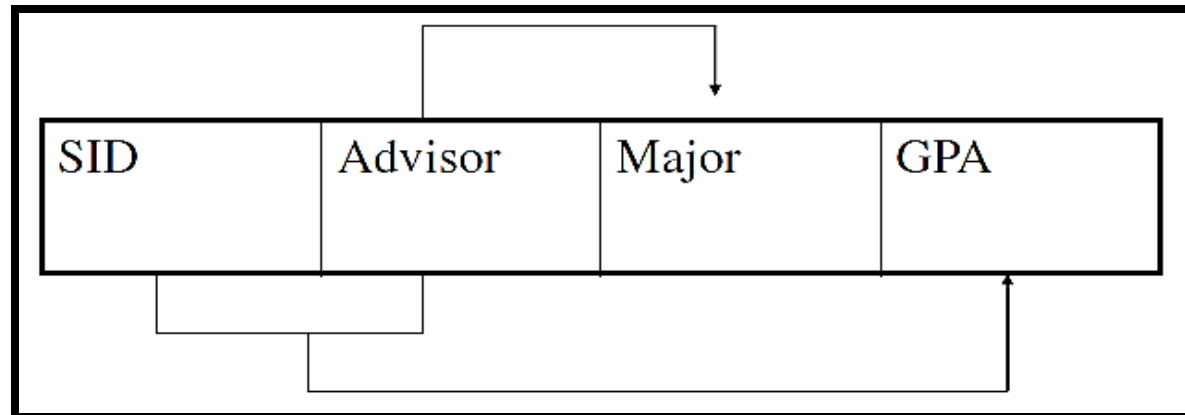
SID	Major	Advisor	GPA
123	Physics	Hawking	4.0
123	Music	Mahler	3.3
456	Literature	Mishener	3.2
789	Music	Bach	3.7
678	Physics	Hawking	3.5

- A given student may have more than one major, for each major a student has exactly one advisor and one GPA.
- A major may have several faculty members as advisers.
- Each advisor advises in exactly one major.

Converting from 3NF to BCNF



SID, Major → GPA, Advisor
Advisor → Major



SID, Advisor → GPA, Major
Advisor → Major

Tables in BCNF

Student			Advisor	
SID	Advisor	GPA	Advisor	Major
123	Hawking	4.0	Hawking	Physics
123	Mahler	3.3	Mahler	Music
456	Mishener	3.2	Mishener	Literature
789	Bach	3.7	Bach	Music
678	Hawking	3.5		

A relation is in BCNF if every determinant is a candidate key.

In BCNF:

SID, Advisor → GPA,

Advisor → Major

Fourth Normal Form (4NF)

An entity type is in 4NF if it is in 3NF/BCNF and there are **no multivalued dependencies** between its attribute types.

Any entity type in 3NF is transformed to 4NF:

- Detect any multivalued dependencies.
- Decompose entity type.

Multi-valued Dependency (MVD):

Represent a dependency between attributes (eg. A, B, C) in a relation, such that for each value of A there is a set of values for B and a set for C. However, the set of values for B and C are independent of each other.

- $A \twoheadrightarrow B$
- $A \twoheadrightarrow C$

Multivalued Dependencies

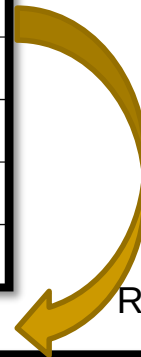
Course	Instructor	Textbook
Management	White	Drucker
	Green	Peters
	Black	
Finance	Gray	Jones Chang

- Each course has a well-defined set of instructor (management has 3 instructors)
- Each course has a well-defined set of textbooks that are used (Finance has 2 textbooks)
- The textbooks that are used for a given course are independent of the instructor of the course (the same two textbooks are used for management regardless of which the three instructors is teaching mgmt)

Converting from BCNF to 4th Normal Form

Course	Instructor	Textbook
Management	White	Drucker
Management	White	Peters
Management	Green	Drucker
Management	Green	Peters
Management	Black	Drucker
Management	Black	Peters
Finance	Gray	Jones
Finance	Gray	Chang

Relation R



Relation R1

Relation R2

Course	Instructor	Course	Textbook
Management	White	Management	Drucker
Management	Green	Management	Peters
Management	Black	Finance	Jones
Finance	Gray	Finance	Chang

An entity type is in 4NF if it is in 3NF/BCNF and there are **no multivalued dependencies** between its attribute types.



Normalization Example (Project_Detail) - Unnormalized Form (UNF)

- *Assume* that Project Budget can be determined by Project Manager.
 - It will be 1 to 1 relationship in BCNF.
- The Project_Detail base relation has the dependency to the following attributes:

Project_Detail (Project_Code, Project_Title, Project_Manager, Project_Budget, Employee_No, Employee_Name, Department_No, Department_Name, Hourly_Rate)

Project_Detail First Normalized Form (1NF)

<u>Project Code</u>	<u>Project Title</u>	<u>Project Manager</u>	<u>Project Budget</u>	<u>Employee No.</u>	<u>Employee Name</u>	<u>Department No.</u>	<u>Department Name</u>	<u>Hourly Rate</u>
PC010	Pensions System	M Phillips	24500	S10001	A Smith	L004	IT	22.00
PC010	Pensions System	M Phillips	24500	S10030	L Jones	L023	Pensions	18.50
PC010	Pensions System	M Phillips	24500	S21010	P Lewis	L004	IT	21.00
PC045	Salaries System	H Martin	17400	S10010	B Jones	L004	IT	21.75
PC045	Salaries System	H Martin	17400	S10001	A Smith	L004	IT	18.00
PC045	Salaries System	H Martin	17400	S31002	T Gilbert	L028	Database	25.50
PC045	Salaries System	H Martin	17400	S13210	W Richards	L008	Salary	17.00
PC064	HR System	K Lewis	12250	S31002	T Gilbert	L028	Database	23.25
PC064	HR System	K Lewis	12250	S21010	P Lewis	L004	IT	17.50
PC064	HR System	K Lewis	12250	S10034	B James	L009	HR	16.50

Normalization Example (Project_Detail)

Unnormalized Form (UNF): A table that contains one or more repeating groups.

- **Project_Detail** (Project Code, Project_Title, Project_Manager, Project_Budget, Employee No, Employee_Name, Department_No, Department_Name, Hourly_Rate)

First Normal Form (1NF): A table without the repeating groups.

Determine the keys to uniquely identify each row.

- **Project_Detail** (Project Code, Project_Title, Project_Manager, Project_Budget, Employee No, Employee_Name, Department_No, Department_Name, Hourly_Rate)

Normalization Example (Project_Detail)

Second Normal Form (2NF):

- Project (Project_code, Project_Title, Project_Manager, Project_Budget)
- Project_Detail (Project Code*, Employee No*, Hourly_Rate)
- Employee (Employee No, Employee_Name, Department_No, Department_Name)

Third Normal Form (3NF):

- Project (Project_code, Project_Title, Project_Manager, Project_Budget)
- Project_Detail (Project Code*, Employee No*, Hourly_Rate)
- Employee (Employee No, Employee_Name, Department_No*)
- Department (Department No, Department_Name)

Normalization Example (Project_Detail)

- Boyce-Codd Normal Form (BCNF)

Boyce-Codd Normal Form (BCNF):

- Assume that Project Budget can be determined by Project Manager:
Every determinant is a candidate key.
- Project (Project_code, Project_Title, Project_Manager*)
- Budget (Project_Manager, Project_Budget)
- Project_Detail (Project_Code, Employee_No, Hourly_Rate)
- Employee (Employee_No, Employee_Name, Department_No*)
- Department (Department_No, Department_Name)

Normalization Example (Project_Detail)

- Dependencies

■ Full Dependencies:

Project_Code, Employee_No $\rightarrow$ Project_Title, Project_Manager,
Project_Budget, Employee_Name, Department_No,
Department_Name, Hourly_Rate

■ Partial Dependencies:

Project_Code, $\rightarrow$ Project_Title, Project_Manager, Project_Budget

Employee_No $\rightarrow$ Employee_Name, Department_No,
Department_Name

■ Transitive Dependencies:

Department_No $\rightarrow$ Department_Name

De-Normalized Designs

When a normalized design is unnatural, awkward, or results in unacceptable performance, a de-normalized design is preferred.

Example:

- Normalized relation:
 - CUSTOMER (CustNumber, CustName, Zip)
 - CODES (Zip, City, State)
- De-Normalized relations:
 - CUSTOMER (CustNumber, CustName, City, State, Zip)

Advantages and Disadvantages of De-Normalization

Advantages - improve performance by:

- Pre-computing derived data.
- Minimizing the need for joins.
- Reducing the number of foreign keys in relations.
- Reducing the number of indexes (thereby saving storage space).
- Reducing the number of relations.

Disadvantages:

- May speed up retrievals but can slow down updates.
- Can increase the size of relations.
- May simplify implementation in some cases but may make it more complex in others.
- Sacrifices flexibility.

Methods of De-normalization

Possible ways:

- Reduce joins.
- Introducing repeating groups.
- Combining 1:1 relationships.
- Merging lookup tables with base relation.
- Creating extract tables.
- Duplicating foreign key attributes in 1:* relationships.

The End Of Lecture 4

Thank You